CLAIMS

What is claimed is:

1. A balance control method for a wheel-legged robot, the method being executed by a computer device, the wheel-legged robot comprising: a moving wheel, n links, and n rotating joints, the moving wheel being connected to a first link of the n links through a first rotating joint of the n rotating joints, and the n links being connected in series through n-1 rotating joints other than the first rotating joint, where n being a positive integer greater than 1; the method comprising:

acquiring a state quantity of the wheel-legged robot at a first moment, the state quantity at the first moment indicating a motion state of the wheel-legged robot at the first moment;

determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment, the dynamics model parameters being configured to define a mapping relationship between an angular acceleration at the first moment and a rotation torque at a second moment, the angular acceleration at the first moment comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second moment comprising rotation torques of the n rotating joints at the second moment;

establishing a sliding surface according to the state quantity at the first moment, the state quantity of the wheel-legged robot approaching a stable value on the sliding surface;

calculating the rotation torques of the n rotating joints according to the sliding surface and the dynamics model parameters; and

controlling the n rotating joints according to the rotation torques of the n rotating joints at the second moment.

2. The method according to claim 1, wherein the determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment comprises:

substituting the state quantity at the first moment into the dynamics equation to determine an inertia matrix, a bias force matrix, and a gravity matrix at the first moment, the inertia matrix indicating masses and moments of inertia of the n rotating joints at the first moment, the bias force matrix indicating a bias force of the wheel-legged robot at the first moment, the gravity matrix indicating a gravity of the wheel-legged robot at the first moment; and

determining the dynamics model parameters according to the inertia matrix, the bias force matrix, and the gravity matrix.

3. The method according to claim 2, wherein the dynamics model parameters comprise: a proportion parameter matrix and an offset parameter matrix, the proportion parameter matrix indicating a proportional relationship between the angular acceleration at the first moment and the rotation torque at the second moment, the offset parameter matrix indicating an offset relationship between the angular acceleration at the first moment and the rotation torque at the second moment;

the determining the dynamics model parameters according to the inertia matrix, the bias force matrix, and the gravity matrix comprises:

respectfully processing, by using a selection matrix, a product of an inverse matrix of the inertia matrix and the bias force matrix, and a product of the inverse matrix of the inertia matrix and the gravity matrix, to obtain the offset parameter matrix, the selection matrix being configured to extract the rotation torques of the n rotating joints from the dynamics equation; and

processing the inverse matrix of the inertia matrix by using the selection matrix to obtain the proportion parameter matrix.

4. The method according to claim 1, wherein the dynamics equation of the robot is obtained by abstracting the wheel-legged robot into an n-level inverted pendulum model and performing derivation based on an Euler-Lagrange equation.

5. The method according to claim 1, wherein the sliding surface comprises n sliding surfaces, and the n sliding surfaces are configured to constrain the rotation torques of the n rotating joints;

the establishing a sliding surface according to the state quantity at the first moment comprises:

determining at least two sliding mode parameters for an ith sliding surface of the n sliding surfaces, i being a positive integer less than or equal to n; and

establishing the ith sliding surface according to the at least two sliding mode parameters and the state quantity at the first moment.

6. The method according to claim 5, wherein the determining at least two sliding mode parameters for an ith sliding surface of the n sliding surfaces comprises:

determining a first sliding mode parameter of the at least two sliding mode parameters from a $(2i-1)^{th}$ predicted parameter set, and determining a second sliding mode parameter of the at least two

sliding mode parameters from a $(2i)^{th}$ predicted parameter set, wherein

sliding mode parameters respectively comprised in the $(2i-1)^{th}$ predicted parameter set and the $(2i)^{th}$ predicted parameter set satisfy a constraint condition of a stability criterion.

7. The method according to claim 5, wherein the state quantity comprises: deflection angles of the n links, angular velocities of the n links, and an angular velocity of the moving wheel;

the establishing the ith sliding surface according to the at least two sliding mode parameters and the state quantity at the first moment comprises:

processing the deflection angle of an ith link according to the first sliding mode parameter of the at least two sliding mode parameters to obtain a processing result of the ith link;

processing the angular velocity of the moving wheel according to the second sliding mode parameter of the at least two sliding mode parameters to obtain a processing result of the moving wheel; and

establishing the ith sliding surface according to the processing result of the ith link, the processing result of the moving wheel, and the angular velocity of the ith link.

8. The method according to claim 1, wherein the controlling the n rotating joints according to the rotation torques of the n rotating joints comprises:

for a rotating joint of the n rotating joints, controlling a rotating motor corresponding to the rotating joint to rotate according to the rotation torque of the rotating joint.

9. The method according to claim 1, wherein the controlling the n rotating joints according to the rotation torques of the n rotating joints comprises:

calculating a task acceleration of the wheel-legged robot at the second moment according to the rotation torques of the n rotating joints, the task acceleration comprising an acceleration related to a mass center of the wheel-legged robot;

determining force and torque instructions for whole body joints of the wheel-legged robot at the second moment according to the task acceleration, the whole body joints comprising the n rotating joints; and

controlling the whole body joints according to the force and torque instructions at the second moment.

10. The method according to claim 9, wherein the calculating a task acceleration of the wheel-legged robot at the second moment according to the rotation torques of the n rotating joints comprises:

determining an angular acceleration at the second moment according to the dynamics equation and the rotation torques of the n rotating joints;

determining an expected incremental position and an expected incremental speed of the mass center of the wheel-legged robot at the second moment according to the state quantity at the first moment and the angular acceleration at the second moment, the expected incremental position indicating a distance between a projection of the mass center of the wheel-legged robot on a contact surface and a virtual contact point in a first direction, the expected incremental speed indicating a changing speed of the distance in the first direction, the virtual contact point referring to a center of contact points between the wheel-legged robot and the contact surface; and

determining the task acceleration of the wheel-legged robot at the second moment according to the expected incremental position and the expected incremental speed.

11. The method according to claim 1, wherein the state quantity comprises: the deflection angles of the n links, the angular velocities of the n links, and the angular velocity of the moving wheel;

the acquiring a state quantity of the wheel-legged robot at a first moment comprises:

determining the deflection angles of the n links through an inertial measurement unit and a motor encoder of the wheel-legged robot; and

determining the angular velocities of the n links and the angular velocity of the moving wheel through the motor encoder.

12. A balance control apparatus for a wheel-legged robot, the wheel-legged robot comprising: a moving wheel, n links, and n rotating joints, the moving wheel being connected to a first link of the n links through a first rotating joint of the n rotating joints, and the n links being connected in series through n-1 rotating joints other than the first rotating joint, where n being a positive integer greater than or equal to 2; the apparatus comprising:

a processor and a memory, the memory having a computer program stored therein, the computer program being loaded and executed by the processor to implement:

acquiring a state quantity of the wheel-legged robot at a first moment, the state quantity at the first moment indicating a motion state of the wheel-legged robot at the first moment;

determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment, the dynamics model parameters being configured to define a mapping relationship between an angular acceleration at the first moment and a rotation torque at a second moment, the angular acceleration at the first moment comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second moment comprising rotation torques of the n rotating joints at the second moment;

establishing a sliding surface according to the state quantity at the first moment, the state quantity of the wheel-legged robot approaching a stable value on the sliding surface;

calculating the rotation torques of the n rotating joints according to the sliding surface and the dynamics model parameters; and

controlling the n rotating joints according to the rotation torques of the n rotating joints at the second moment.

13. A non-transitory computer-readable storage medium, the computer-readable storage medium having a computer program stored therein, the computer program being loaded and executed by a processor, causing the processor to implement:

acquiring a state quantity of the wheel-legged robot at a first moment, the state quantity at the first moment indicating a motion state of the wheel-legged robot at the first moment;

determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment, the dynamics model parameters being configured to define a mapping relationship between an angular acceleration at the first moment and a rotation torque at a second moment, the angular acceleration at the first moment comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second moment comprising rotation torques of the n rotating joints at the second moment;

establishing a sliding surface according to the state quantity at the first moment, the state quantity of the wheel-legged robot approaching a stable value on the sliding surface;

calculating the rotation torques of the n rotating joints according to the sliding surface and the dynamics model parameters; and

controlling the n rotating joints according to the rotation torques of the n rotating joints at the second moment.

14. The storage medium according to claim 13, wherein the determining dynamics model parameters according to a dynamics equation of the wheel-legged robot and the state quantity at the first moment comprises:

substituting the state quantity at the first moment into the dynamics equation to determine an inertia matrix, a bias force matrix, and a gravity matrix at the first moment, the inertia matrix indicating masses and moments of inertia of the n rotating joints at the first moment, the bias force matrix indicating a bias force of the wheel-legged robot at the first moment, the gravity matrix indicating a gravity of the wheel-legged robot at the first moment; and

determining the dynamics model parameters according to the inertia matrix, the bias force matrix, and the gravity matrix.

15. The storage medium according to claim 14, wherein the dynamics model parameters comprise: a proportion parameter matrix and an offset parameter matrix, the proportion parameter matrix indicating a proportional relationship between the angular acceleration at the first moment and the rotation torque at the second moment, the offset parameter matrix indicating an offset relationship between the angular acceleration at the first moment and the rotation torque at the second moment;

the determining the dynamics model parameters according to the inertia matrix, the bias force matrix, and the gravity matrix comprises:

respectfully processing, by using a selection matrix, a product of an inverse matrix of the inertia matrix and the bias force matrix, and a product of the inverse matrix of the inertia matrix and the gravity matrix, to obtain the offset parameter matrix, the selection matrix being configured to extract the rotation torques of the n rotating joints from the dynamics equation; and

processing the inverse matrix of the inertia matrix by using the selection matrix to obtain the proportion parameter matrix.

16. The storage medium according to claim 13, wherein the dynamics equation of the robot is obtained by abstracting the wheel-legged robot into an n-level inverted pendulum model and performing derivation based on an Euler-Lagrange equation.

17. The storage medium according to claim 13, wherein the sliding surface comprises n sliding surfaces, and the n sliding surfaces are configured to constrain the rotation torques of the n rotating joints;

the establishing a sliding surface according to the state quantity at the first moment comprises:

determining at least two sliding mode parameters for an ith sliding surface of the n sliding surfaces, i being a positive integer less than or equal to n; and

establishing the ith sliding surface according to the at least two sliding mode parameters and the state quantity at the first moment.

18. The storage medium according to claim 17, wherein the determining at least two sliding mode parameters for an ith sliding surface of the n sliding surfaces comprises:

determining a first sliding mode parameter of the at least two sliding mode parameters from a $(2i-1)^{th}$ predicted parameter set, and determining a second sliding mode parameter of the at least two sliding mode parameters from a $(2i)^{th}$ predicted parameter set, wherein

sliding mode parameters respectively comprised in the $(2i-1)^{th}$ predicted parameter set and the $(2i)^{th}$ predicted parameter set satisfy a constraint condition of a stability criterion.

19. The storage medium according to claim 17, wherein the state quantity comprises: deflection angles of the n links, angular velocities of the n links, and an angular velocity of the moving wheel;

the establishing the ith sliding surface according to the at least two sliding mode parameters and the state quantity at the first moment comprises:

processing the deflection angle of an ith link according to the first sliding mode parameter of the at least two sliding mode parameters to obtain a processing result of the ith link;

processing the angular velocity of the moving wheel according to the second sliding mode parameter of the at least two sliding mode parameters to obtain a processing result of the moving wheel; and

establishing the ith sliding surface according to the processing result of the ith link, the processing result of the moving wheel, and the angular velocity of the ith link.

20. The storage medium according to claim 13, wherein the controlling the n rotating joints according to the rotation torques of the n rotating joints comprises:

for a rotating joint of the n rotating joints, controlling a rotating motor corresponding to the rotating joint to rotate according to the rotation torque of the rotating joint.